GUQINAGENT: Tool-Interactive Performance-Action Planning for Jianpu-to-Jianzipu Generation

Anonymous ACL submission

Abstract

Jianpu specifies pitches and rhythm, whereas jianzipu describes the actions used to perform them on the guqin. Generating jianzipu requires choosing among multiple realizations whose pitches and execution depend on tuning, inherited notation state, and the surrounding phrase. We introduce GUQINAGENT, a framework that learns tool-guided generation from expert-annotated scores. An offline teacher constructs executable interaction trajectories, from which public supervision trains single-stage and two-stage agents. The full two-stage system separates basic fingering from idiomatic refinement; both variants use pitch candidates, editable plans, and diagnostic feedback. On 48 scores, the full system achieves 99.981% pitch accuracy within 50 cents on auditor-resolvable events, compared with 86.237% without diagnostic feedback. Our single-stage variant achieves 98.732% and higher reference fingering and ornament agreement. Modern-melody results further show that conditional pitch accuracy and verified-correct output coverage can favor different variants. Evaluation therefore combines pitch fidelity, coverage, reference-action agreement, and runtime diagnostics, with performer assessment planned for feasibility and musical expression. We will release the model, redacted training trajectories, benchmark, and runtime.

1 Introduction

A melody specifies which notes to play, but realizing it on the guqin also requires deciding how to produce and connect them. Jianzipu records these performance actions through combinations of strings, hui positions, left-hand fingers, right-hand strokes, and techniques (Li et al., 2026b,a). Preparing a jianzipu realization from a jianpu score therefore calls for instrument-specific judgment. A system that proposes editable realizations and exposes detectable errors could support performers and learners in arranging melodies for the guqin.

The goal involves four related requirements: pitch fidelity, physical playability, adherence to playing conventions, and idiomatic musical expression. Several string–position combinations may produce the same pitch, yet their hand shapes, transitions, and tonal effects can differ. Two individually valid notes may be difficult to execute together or connect naturally. Suitable choices of slides, vibrato, and compound gestures also contribute to *yunwei*, the guqin's characteristic expressive quality. Adding more ornaments alone does not establish this quality; their placement and relation to the phrase matter.

These requirements create three challenges. First, generation involves choosing among multiple realizations under both physical and musical constraints. Pitch agreement does not establish whether a hand position is feasible, whether a stroke follows conventional usage, or whether a technique suits the phrase. Second, decisions depend on the evolving state of the complete score. A harmonic region remains active until terminated, and subsequent notation may omit positions inherited from earlier events. The same string and hui can therefore imply different pitches inside and outside a harmonic region. Local edits must be interpreted in context and may change the meaning of later entries.

Third, expert annotations provide final realizations but generally omit the process of choosing, checking, and revising them. Final notation alone does not demonstrate when an agent should query pitch candidates, inspect preceding material, or respond to a warning. Recent jianzipu research addresses glyph recognition and structured extraction from manuscript pages (Li et al., 2026b,a). Here, the input is a complete machine-readable jianpu score, and the task is to construct a guqin performance realization. This setting connects symbolic music generation with planning actions whose interpretation and feasibility depend on prior choices.

We address these challenges through interactive plan editing. Following work on reasoning with tools and iterative feedback (Yao et al., 2023; Schick et al., 2023; Madaan et al., 2023), GUQINAGENTpartitions a complete score into manageable work units while maintaining one shared notation plan. Its runtime provides tuning-aware candidates and conservative checks for pitch, state, and supported performance rules. A Fingering stage establishes a basic realization; a Guqinizer stage considers idiomatic technique and continuity. An offline teacher uses private expert annotations to construct actual tool interactions, which are redacted into student-visible supervision. Automatic checks support the verifiable parts of the task, while performer assessment is needed to judge expressive suitability. In summary, our contributions are fourfold:

1. We introduce *constrained jianzipu generation*, a benchmark and executable runtime for reproducible score-to-jianzipu evaluation.

2. We introduce warning-guided inference: deterministic pitch and conservative notation-constraint diagnostics become visible, actionable feedback rather than hidden post-hoc validation.

3. We design GUQINAGENT, with a single-stage trajectory-trained variant and a full two-stage Fingering→Guqinizer agent that distinguishes basic sound production from stylistic refinement. Their comparison measures pitch consistency and reference-action agreement; performer evaluation addresses execution and expression.

4. We develop targeted inverse trajectory construction, which replays annotations through the runtime and exports public, redacted supervision.

2 Related Work

Guqin notation and performance-oriented symbolic music. Recent work on jianzipu has focused on recovering structured notation from glyphs and manuscript pages (Li et al., 2026b,a). Earlier language-model-based work also studied transcription from jianzipu into modern notation (Shi et al., 2024). Our setting reverses this direction: starting from melodic events, the system must choose how to realize them on the instrument. A related problem arises in guitar tablature inference, where a pitch may admit several string–fret assignments and playability is assessed by musicians (Edwards et al., 2024). Work on piano fingering likewise models hand position and execution difficulty (Iman and Ahn, 2024). At the composition level, YuE2 generates a readable melody-and-chord score before synthesizing full-song audio; the score supports direct editing and revisions driven by external language models in response to user feedback (Yuan et al., 2026). GUQINAGENTstudies how a given score is realized through guqin performance actions, including state-dependent gestures and compact notation whose interpretation depends on tuning and prior actions.

Tool-interactive planning and revision. ReAct interleaves reasoning and environment actions, while Toolformer and ToolLLM investigate tool-use learning and data construction (Yao et al., 2023; Schick et al., 2023; Qin et al., 2024). Self-Refine and Reflexion use iterative feedback to improve model behavior; CRITIC specifically studies revision driven by external tool outputs (Madaan et al., 2023; Shinn et al., 2023; Gou et al., 2024). GUQINAGENTuses a shared executable notation state to check pitch realizations and issue conservative, event-level warnings. These checks inform subsequent edits but do not determine musical suitability on their own.

Learning from interaction trajectories. Agent-Tuning and FireAct demonstrate the value of supervised interaction trajectories for language agents (Zeng et al., 2024; Chen et al., 2023). More closely related, RetroGen reconstructs process supervision from final expert artifacts for evidence-grounded generation (Huang et al., 2026). Our teacher likewise has access to a target artifact during offline construction, but is required to execute edits in the student-facing guqin runtime; the retained training context contains only public actions and observations. GEPA instead optimizes prompts by reflecting on execution feedback without updating model weights, providing a distinct comparison to trajectory fine-tuning (Agrawal et al., 2026).

3 Methodology

3.1 Constrained Generation with an Executable Runtime

Given a complete jianpu score $x = (m, e_{1:N})$, the task is to construct an event-indexed jianzipu plan $y = (y_1, \ldots, y_N)$. The metadata m specifies the

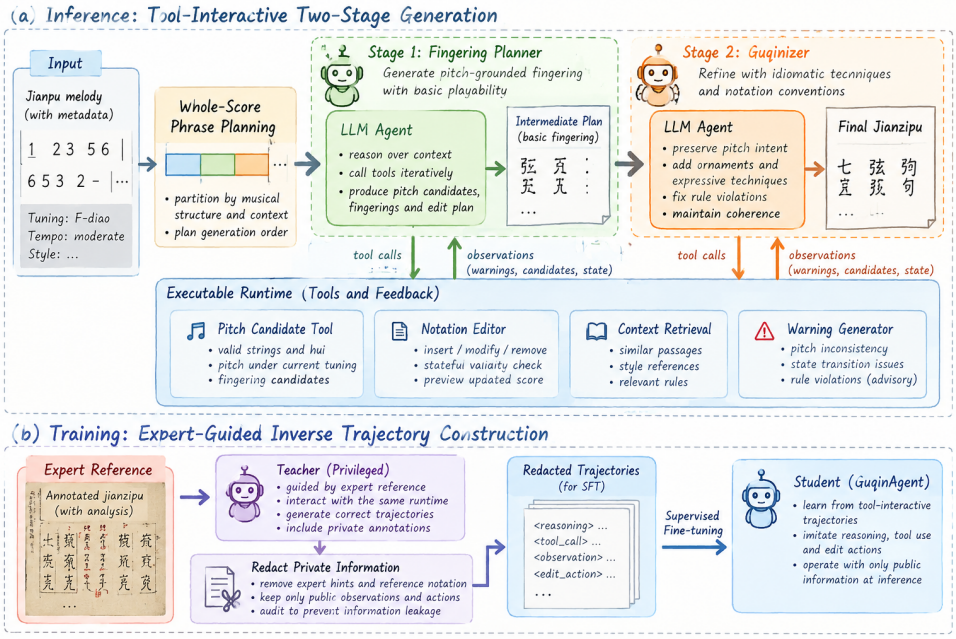


Figure 1: Overview of GUQINAGENT. **Top:** at inference time, the agent partitions a complete jianpu score into ordered work units, constructs a pitch-grounded intermediate plan in the Fingering stage, and refines it into final jianzipu in the Guqinizer stage. Both stages interact with the same executable runtime, which returns pitch candidates, plan previews, compact context, and advisory warnings. **Bottom:** for training, a privileged teacher uses expert annotations only through a private channel while interacting with the public runtime. Redaction retains a student-visible trajectory for supervised fine-tuning, so the deployed student conditions only on score context, its prior actions, and runtime observations.

mode and tuning; each event e_i has a stable index, notated duration, structural type, and a target pitch q_i when defined. An output entry y_i is either a jianzipu expression or an explicit empty display value. The latter may indicate continuation or coverage by a compound gesture and does not remove the event from the score. ABC notation is used only as an internal representation for pitch processing, not as an additional input.

Unlike independent token prediction, the interpretation of y_i depends on preceding actions. We represent this dependence by a deterministic notation-state transition,

$$s_i = \mathcal{T}(s_{i-1}, y_i; m), \qquad s_0 = s_{\text{init}}(m), \tag{1}$$

where s_i records such properties as harmonic-region status, inherited left-hand position, walking gestures, and compound-action coverage. Given this state, the realized pitch $\hat{q}_i = \mathcal{P}(y_i, s_{i-1}; m)$ is a partial function: some compound or ambiguous expressions do not admit a reliable pitch estimate. Structural consistency and pitch agreement are therefore distinct from the broader question of musical suitability.

Interactive plan editing. The complete score is partitioned into an ordered sequence of bounded, musically aligned work units $\mathcal{B}(x) = (p_1, \ldots, p_K)$. This partition is computed from the input score; the benchmark examples remain complete scores. The agent edits one shared plan rather than generating independent phrases. At interaction step t within p_k, its public history consists of the score, current work unit, editable plan, replayed notation state, and previous tool observations:

$$\begin{aligned} h_t &= (x, p_k, y^{(t)}, s^{(t)}, o_{<t}), \\ a_t &\sim \pi_\theta(\cdot \mid h_t, r), \end{aligned} \tag{2}$$

where r denotes the current planning stage and $s^{(t)}$ is obtained by replaying the current plan. The executable runtime $\mathcal{R}$ processes an action and returns the resulting plan, state, and tool observation:

$$(y^{(t+1)}, s^{(t+1)}, o_t) = \mathcal{R}(x, y^{(t)}, s^{(t)}, a_t). \tag{3}$$

Read-only actions leave the plan unchanged. The runtime used to construct training trajectories is also used at inference time.

The agent can request tuning-aware string–hui realizations through `get_pitch_candidates`, including open strings, stopped notes, harmonics,

and supported inherited or compound gestures. Returned candidates distinguish exact, approximate, and unavailable matches. The only plan-writing operation is `edit_plan(jianzi_rows)`, which applies event-indexed updates and replays the resulting notation state. Compact context retrieval through `list_context` and targeted earlier-phrase lookup provides prior material without reproducing the entire score in every interaction.

Conservative constraint feedback. We distinguish runtime-checkable violations from cases whose interpretation is uncertain. For a rule c and event i, let $v_c(i; x, y, s) \in \{0, 1, \perp\}$ denote no detected violation, a detected violation, or an indeterminate check, respectively. The warning set is

$$W(x, y) = \{(i, c) : v_c(i; x, y, s) = 1\}, \quad (4)$$

with s obtained by replaying y. In the implementation, checks cover reliably parsed pitch mismatches and conservative diagnostics for harmonic-state continuity, walk direction or span, interrupted open-string resonance, paired stroke adjacency, and difficult left-hand reaches. Indeterminate cases are not converted into violations. Warnings accompany the relevant `edit_plan` preview but neither rewrite the plan nor imply that an unflagged output is musically correct. The agent may query candidates, revise the affected entries, or retain its choice; identical repeated edits are stopped to prevent unproductive loops.

3.2 Two-Stage Performance-Action Planning

The same notation event jointly specifies how a pitch is produced and how it is articulated. We separate these decisions into two successive planning stages. *Fingering* first constructs a basic pitch-grounded realization, choosing strings, hui positions, left-hand contact, and elementary right-hand strokes. *Guqinizer* then revises the committed plan for idiomatic execution, including ornaments, walking gestures, harmonic transitions, and compound-action notation. The second stage may preserve or revise earlier decisions; it is not a post-processing formatter.

For work unit p_k, let $y^{(k-1)}$ be the shared plan after completing the preceding units. A bounded runtime rollout under each stage gives

$$\begin{aligned} y^{F,(k)} &= \mathrm{Rollout}_F(x, p_k, y^{(k-1)}; \mathcal{R}), \\ y^{(k)} &= \mathrm{Rollout}_G(x, p_k, y^{F,(k)}; \mathcal{R}), \end{aligned} \quad (5)$$

where each rollout consists of stage-conditioned tool interactions of the form in Eq. (3). Only edits committed through the runtime enter the shared plan. Crucially, the state passed to p_{k+1} is replayed from $y^{(k)}$, including any earlier mistakes, rather than substituted from an expert reference. Both stages retain access to pitch candidates and runtime feedback. Warnings can trigger another revision but are not hard acceptance criteria; execution remains subject to the configured interaction budget.

This decomposition imposes a procedural bias: establish a basic sound-producing realization before making the more context-dependent choices involved in idiomatic execution. It does not assume a unique separation between fingering and ornamentation, nor does it guarantee that the second stage preserves every correct decision. We evaluate both the benefits and regressions introduced by refinement against a single-stage agent using the same public tools.

3.3 Expert-Guided Inverse Trajectory Learning

Expert jianzipu annotations specify desired performance actions but generally not the tool interactions by which an agent could arrive at them. Directly supervising final notation therefore leaves the interaction policy unspecified. We instead construct trajectories by allowing an offline teacher to work backward from the annotated target while acting through the same executable runtime available to the student.

Let z be a phrase-level expert annotation and $\mathcal{K}(z)$ the technique-relevant descriptions retrieved from a curated knowledge base. Both are private to the teacher. Its rollout is conditioned on the input, private guidance, and public runtime:

$$\tau^T \sim \mathrm{Rollout}\big(\pi_T \mid x, z, \mathcal{K}(z); \mathcal{R}\big), \quad (6)$$

where τ^T records the chronological sequence of tool actions, returned observations, and accepted plan updates. The reference guides the search for an executable realization but does not constitute a tool response. The teacher constructs stage-specific traces: basic fingering first, followed by idiomatic refinement. Semantic descriptions clarify technique constraints that may be implicit in compact jianzipu symbols; they are not added to the student's inference context.

Validation and public-prefix reconstruction. Candidate examples are screened for reference pars-

ing and consistency before construction, and generated traces are checked against the runtime. Training examples are obtained by removing private expert content and retaining the public interaction sequence,

$$\tilde{\tau} = g_{\text{pub}}(\tau^T), \quad \mathcal{D} = \{\tilde{\tau} : \tau^T \text{ passes validation}\}. \tag{7}$$

The intended information boundary is temporal as well as structural: at each supervised action, the conditioning context should contain only the input, earlier public actions, and observations already returned by the runtime. Removing private fields alone cannot establish this property if rewritten reasoning anticipates future observations or reproduces unseen annotation details. Public-prefix consistency must therefore be checked separately from runtime execution validity; the precise checks and filtering counts belong in the experimental protocol.

Student supervision. We fit a stage-conditioned student policy to the assistant-side actions and messages in the retained public trajectories. For a supervised token position u, let h_u^{pub} contain the preceding student-visible context and let w_u be its target token. The training loss is standard masked cross-entropy,

$$\mathcal{L}_{\text{SFT}}(\theta) = -\sum_{\tilde{\tau}\in\mathcal{D}} \sum_{u\in\mathcal{Y}(\tilde{\tau})} \log p_\theta(w_u \mid h_u^{\text{pub}}), \tag{8}$$

where $\mathcal{Y}(\tilde{\tau})$ indexes supervised assistant tokens; tool outputs form context but are not prediction targets. At deployment, neither expert annotations nor private semantic guidance are provided. The resulting student must recover its actions from the input score, its own previous edits, and the executable feedback it has observed.

4 Experiments

4.1 Experimental Setup

Benchmark and Data Protocol. The test set contains 48 complete scores, partitioned into 1,031 work units, with 23,602 aligned event records. Scores are the units of dataset splitting, while phrases are internal work units used by the agent. The single-stage student and both two-stage SFT conditions produce protocol-valid outputs for all 1,031 work units.

Component	Contents
Input	Complete-score metadata, tuning, jianpu events, and stable event indices.
Output	One event-indexed jianzipu plan for the complete score.
Runtime	Pitch candidates, editing, context retrieval, validation, and diagnostic feedback.
Reference	Expert notation for offline supervision and structured action comparison.

Table 1: Benchmark representation. Reference notation is withheld from inference.

All outputs are evaluated using the same current rules after replaying each complete score in chronological order. This preserves harmonic regions, left-hand positions, and other state across work-unit boundaries. Runtime snapshots and evaluation code fingerprints are retained for auditability.

Student Training and Inference Budgets. The training split contains 153 scores and 3,976 work units. Two-stage supervision yields 7,952 stage-specific conversations, jointly training one adapter used by both stages; single-stage trajectory supervision, fixed-workflow supervision, and Direct SFT each yield 3,976 examples. All four students start from Qwen3.5-9B and use 4-bit LoRA fine-tuning for three epochs, with rank 8, scaling $\alpha = 16$, learning rate 2×10^{-4}, effective batch size 12, and an 8,192-token training cutoff. Equal epochs do not imply equal numbers of conversations or supervised tokens. Standard preprocessing truncation applies at the cutoff; counting a conversation does not certify that every original token is supervised.

At inference, local Qwen conditions use a 16,384-token context and at most 8,192 output tokens per request. Tool-based single-stage conditions allow up to 16 rounds per attempt; two-stage conditions allow eight rounds per stage attempt, with at most three attempts and, when feedback is enabled, three Guqinizer passes. These are ceilings, not required numbers of interactions. Feedback OFF performs one Guqinizer pass. Direct generates once per work unit, without tools or repair. Thus these are deployed-pipeline comparisons, not equal-total-token experiments. Full configurations, sampling settings, and dataset/checkpoint fingerprints appear in Appendix A.

Evaluation Metrics. Pitch correctness is measured against the input jianpu melody. The expert jianzipu annotation supplies reference actions for fingering and technique metrics. Let $\mathcal{I}$ contain

Method	Valid coverage ↑	Pitch Acc. ↑	MAE ↓	String–Hui F1 ↑	Tech. Sim. ↑	Orn. F1 ↑	Rule Violation Rate ↓
(a) Traditional scores: 48 scores, 1,031 work units							
Original Qwen3.5-9B, two-stage agent[†]	1.07	N/A	N/A	N/A	N/A	N/A	N/A
GLM-5.3, ReAct + Handbook	100.00	99.760	2.542	30.927	0.6323	7.158	5.044
GLM API, agent + GEPA	–	–	–	–	–	–	–
Fixed-workflow SFT, single-stage	100.00	26.778	517.624	11.002	**0.8401**	**27.928**	95.829
GUQINAGENT, single-stage	100.00	98.732	7.880	**37.511**	0.7618	20.929	14.258
GUQINAGENT w/o feedback	100.00	86.237	64.993	31.678	0.7616	18.339	78.468
GUQINAGENT (Ours)	100.00	**99.981**	**2.308**	35.444	0.7301	17.013	**0.194**
(b) Modern melodies: 10 scores, 91 work units							
GLM-5.3, ReAct + Handbook	100.00	97.699	16.035	36.627	0.7118	39.961	5.495
GLM API, agent + GEPA	–	–	–	–	–	–	–
Fixed-workflow SFT, single-stage	100.00	19.298	558.105	9.409	0.7171	39.084	96.703
GUQINAGENT, single-stage	100.00	97.932	11.999	43.490	0.8358	**43.445**	18.681
GUQINAGENT w/o feedback	100.00	73.485	111.442	38.099	**0.8623**	42.225	85.714
GUQINAGENT (Ours)	100.00	**98.951**	**10.714**	**45.261**	0.8154	41.729	**3.297**

Table 2: Traditional-score and modern-melody results, shown as upper and lower panels. Coverage, accuracy, F1, and rule violation rate are percentages; MAE is in cents. Tech. Sim. is technique distribution similarity on $[0, 1]$. Valid coverage counts protocol-valid outputs over all work units in the panel; rule violation rate is the fraction of work units with at least one runtime-detected rule violation in the final plan. Pitch accuracy is conditional on auditor-resolvable events. String–Hui F1 compares exact event-local string/contact pairs, including resolved compound positions; reference-position coverage is reported below. Modern reference-action metrics use source-score annotations withheld from inference and measure agreement with one realization. [†]The original-model agent produced valid outputs for 11/59 attempted traditional-score units; 972 were skipped after predecessor failures. N/A denotes unavailable full-test metrics; – denotes pending experiments. Only the full two-stage system is marked Ours; its trained variants are comparisons and ablations. Detailed metrics appear in Appendix B.

events for which the auditor returns a determinate matched or mismatched prediction. Let $a_i = 1$ if event i matches the target within 50 cents under the supported matching rules, and $a_i = 0$ otherwise. Let $\mathcal{A}_i$ contain the tone pairs aligned by the auditor, with absolute error δ_{ij} in cents for pair j. We report

$$\begin{aligned} \mathrm{PitchAcc}_{50} &= \frac{\sum_{i \in \mathcal{I}} a_i}{|\mathcal{I}|}, \\ \mathrm{PitchMAE} &= \frac{\sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{A}_i} \delta_{ij}}{\sum_{i \in \mathcal{I}} |\mathcal{A}_i|}. \end{aligned} \tag{9}$$

Accuracy uses events as its denominator; MAE averages aligned tone pairs, and a compound event can contribute multiple pairs. Unresolved cases are excluded from pitch scoring and must be reported separately. We report each condition's evaluable event count and accuracy on their intersection, since generated notation affects which expressions the auditor resolves. The complete event inventory includes structural and continuation events and is not the pitch denominator.

String–Hui F1 is a micro set F1 over exact $(\text{string}, \text{hui})$ pairs within each event, using the stateful auditor's resolved positions. All resolved positions of a compound event are included, without imposing a primary/secondary ordering. Open strings receive an explicit OPEN contact label; numeric hui positions are normalized to six decimal places. Events with unresolved reference positions are excluded, whereas unresolved predictions at scorable reference events contribute missing pairs. The common reference coverage is 12,327/18,128 target events (68.000%) for traditional scores and 1,280/1,528 (83.770%) for modern melodies; the reference contains 12,821 and 1,480 scorable position pairs, respectively. This is conditional reference-position agreement, not a complete transcription score. Left-hand fingering, right-hand fingering, and tone type are reported in Appendix B. Tone type distinguishes open, stopped, and harmonic notes. Ornament precision, recall, and F1 compare event-level sets of named embellishment and compound actions. They share the technique-distribution lexicon and alias normalization below, but exclude the eight basic plucks (mo, tiao, gou, ti, da, zhai, tuo, and bo). Known compounds remain atomic labels. Harmonic-region start/end markers are retained only as state controls, never as techniques or ornaments. A nonempty dis-

joint label set scores zero F1; two empty corpus sets are undefined. These metrics measure agreement with one expert realization, not correctness of execution.

Technique distribution similarity summarizes the relative use of named performance techniques. Each technique is counted at most once per event, then counts are normalized over all technique incidences in the corpus. For predicted and reference distributions p and q over their joint technique vocabulary $\mathcal{V}$, we use

$$\mathrm{TechSim}(p, q) = 1 - \frac{1}{2} \sum_{t \in \mathcal{V}} |p(t) - q(t)|. \quad (10)$$

The score lies in $[0, 1]$, with higher values indicating closer distributions. This metric excludes notation-state markers such as harmonic-region boundaries. It does not evaluate event order, and identical distributions can assign techniques to different events; ornament F1 retains a local comparison. Tone-type distribution similarity uses the same formula over open, stopped, and harmonic notes and is reported in the appendix.

Rule Violation Rate measures the fraction of evaluated work units whose final plans contain at least one runtime-detected rule violation. We replay complete scores through the same rule checker, including for conditions that hide diagnostic feedback during inference; a final runtime warning is the operational indicator of a detected violation. Each work unit contributes at most one to the numerator, regardless of how many rules it violates. The denominators are 1,031 work units for the traditional test and 91 for the modern test in the complete-coverage comparisons. We separately report the total number of detected violations. Thus this metric measures final-plan rule compliance, not the frequency of warning messages emitted during inference. It covers only the implemented checks: unresolved expressions are not automatically violations, and absence of detected violations does not certify playability or musical quality.

Models and Comparisons. The feedback comparison uses Qwen3.5-9B with the same project-trained two-stage LoRA weights in both conditions. The full system exposes warnings, diagnostic hints, and warning-triggered follow-up interactions. The diagnostic-feedback-off condition hides this feedback and disables its follow-up mechanism while retaining structural validation and pitch candidates. Its public user and tool messages were checked for diagnostic-feedback leakage.

GUQINAGENT, single-stage is our trajectory-trained variant without the Fingering–Guqinizer decomposition. It uses a separately trained Qwen3.5-9B LoRA and generates final jianzipu through the same public tools and diagnostic feedback as the full system. It is allowed up to 16 rounds per attempt, while the full pipeline allows eight rounds per stage attempt and up to three Guqinizer passes. Retries and additional passes make their aggregate budgets different; this is a comparison of the two deployed pipelines. Both use complete model-generated histories and the same scoring rules. In Table 2, our single-stage and feedback-off variants are grouped below the fixed-workflow comparison; Ours highlights the full two-stage configuration.

Direct final-artifact supervision. We additionally train a Qwen3.5-9B Direct SFT comparison using the same 3,976 training examples and final committed plans as our single-stage trajectory supervision. Each example retains the initial public musical context and one assistant JSON target, without intermediate reasoning, tool calls, or tool observations. Inference generates one final plan per work unit without tools or diagnostic revision, using the model's own successful preceding outputs as history. This comparison evaluates the end-to-end artifact-only workflow against trajectory-trained tool interaction; it changes both supervision and inference access to tools. Training is complete; evaluation results remain pending.

We also evaluated the original Qwen3.5-9B checkpoint without the project LoRA using the two-stage agent and the same tool definitions. All 59 attempted inputs matched the test-input hashes. The original-model run allowed 8,192 output tokens per request, eight rounds per stage attempt, up to three attempts, and up to three Guqinizer passes, with a context limit of 16,384 tokens. After a work unit failed, the evaluator skipped the remainder of that score, preserving the requirement that history contain model-generated predecessors. These limits match the current SFT evaluation entry point.

The commercial-model ReAct + Handbook baseline uses GLM-5.3 through its API with the production single-stage prompt and a fixed 73-entry technique handbook. It allows eight rounds and two attempts at temperature 0.25, uses the same tools and scoring pipeline, and produces protocol-valid outputs for all 1,031 units. A separate GLM

API baseline with prompts optimized by GEPA on development data remains pending (Agrawal et al., 2026). These comparisons involve different model backends and do not isolate prompt optimization from model capability. GEPA's reflection model, API settings, optimization budget, and inference costs will be reported. Direct generation with the original Qwen checkpoint remains an optional additional comparison.

4.2 Main Results

Original-model completion and protocol failures. The original-model agent attempted 59 of 1,031 work units across the 48 scores. Eleven attempts produced protocol-valid outputs (18.64% of attempted units), while 48 failed. The remaining 972 units were skipped because a predecessor failed. Valid-output coverage was therefore 11/1,031 (1.07%); skipped units are excluded from the attempted-unit protocol-failure denominator. All three adapted conditions produced valid outputs for all 1,031 units. We do not report full-test musical metrics for the original model or compare its 11 successful units directly with the adapted model's complete-test scores.

Of the 48 failed units, 13 failed during Fingering and 35 during Guqinizer. Logs record output truncation in 45 failed units, at least one attempt reaching round eight in 26, and three consecutive pitch-candidate-query rounds in 31. These observations overlap and are not a mutually exclusive attribution of failures. They suggest that sustained tool interaction and termination behavior are substantial obstacles for the original model under the current protocol, alongside the musical decisions required by the task.

Technique distributions and the commercial-model baseline. Technique distribution similarity is 0.7618 for single-stage SFT, 0.7616 for two-stage feedback OFF, and 0.7301 for the two-stage full system. Thus, fewer runtime warnings do not imply closer aggregate technique usage. GLM-5.3 with ReAct + Handbook attains 99.760% pitch accuracy and 2.542 cents MAE, with 102 final warnings across 52 units (5.044%). Its technique similarity is 0.6323 and ornament F1 is 7.158, below both trained students. Its tone-type distribution similarity is higher, however (Appendix B). These measures capture different aspects of realization; the cross-model comparison alone does not establish that trajectory training is necessary for idiomatic playing.

Diagnostic	ON	OFF
Evaluable pitch events	15,758	15,360
Aligned tone pairs for MAE	16,335	15,958
Common evaluable events	14,289	14,289
Pitch Acc. on common events (%)	99.979	86.941
Final runtime warnings	2	3,436
Work units with warnings	2	809
Rule violation rate (%)	0.194	78.468

Table 3: Diagnostic-feedback comparison. Counts refer to final plans, rescored under the same runtime with complete histories. Each condition covers 1,031 work units. OFF warnings are measured during evaluation despite being hidden during inference.

4.3 Ablation Studies

Pitch correctness and diagnostic feedback. The full system attains 99.981% pitch accuracy on 15,758 evaluable events, compared with 86.237% on 15,360 events when diagnostic feedback is disabled (Table 2). MAE is 2.308 versus 64.993 cents. Since the evaluable sets differ, Table 3 additionally reports accuracy on 14,289 events resolved in both conditions: 99.979% versus 86.941%, a gap of 13.038 percentage points. The observed accuracy gap therefore persists on the same input events. The result concerns auditor-supported realizations and does not establish accuracy on unresolved notation.

Residual warnings and error analysis. The full system leaves two final runtime warnings in two work units, compared with 3,436 warnings across 809 work units for OFF. The rule violation rate is 0.194% versus 78.468%. One remaining ON warning in *Youlan* is an octave error, realizing E6 for a target E7. These cases warrant inspection of inherited state, compound notation, and ordinary pitch choices. Final-warning counts and pitch-error counts use different eligibility and matching rules and should not be treated as interchangeable metrics.

Fingering agreement and idiomatic realization. The full system's string–hui, left-hand, right-hand, and tone-type F1 scores are 35.444, 57.119, 46.009, and 62.543, respectively. Their modest differences from OFF accompany a much larger pitch-accuracy gap. The full system also has lower ornament F1: 17.013 versus 18.339. The current comparison provides evidence of improved pitch consistency without establishing an advantage in ornament selection.

Diagnostic	Single-stage	Two-stage
Evaluable pitch events	15,935	15,758
Common evaluable events	14,580	14,580
Common-event Pitch Acc. (%)	98.807	99.979
Final runtime warnings	381	2
Work units with warnings	147	2
Rule violation rate (%)	14.258	0.194

Table 4: Single-stage and two-stage full-system comparison on the same 48 scores, scored after replaying complete histories. Both achieve 100% valid-output coverage. Different retry and pass structures preclude interpreting this as a strictly equal-total-budget architecture ablation.

Diagnostic	Fixed	Teacher
Evaluable pitch events	13,556	15,935
Common evaluable events	12,792	12,792
Common-event Pitch Acc. (%)	26.931	98.781
Final runtime warnings	11,596	381
Work units with warnings	988	147
Warning rate (%)	95.829	14.258

Table 5: Single-stage trajectory-construction comparison. Fixed uses a scripted query-and-edit sequence; Teacher uses expert-guided interactive trajectories. Both produce valid outputs for all 1,031 units. Some training prompts also differ between the recipes.

Tone distributions indicate that pitch accuracy leaves room for different realization strategies. Among recognized tone types on reference-labeled events, harmonics account for 7.51% of ON predictions, compared with 29.34% of reference tone types. A system can realize the correct pitch through different strings, positions, or tone types, and can also underuse stylistically appropriate techniques. Both possibilities motivate examining action distributions and consulting performers when assessing idiomatic quality.

Our single-stage and two-stage variants. The single-stage student completes all 1,031 work units, with 98.732% pitch accuracy on 15,935 evaluable events and 7.880 cents MAE. The two-stage full system reaches 99.981% on 15,758 events and 2.308 cents MAE. On their 14,580 common evaluable events, accuracy is 98.807% for single-stage and 99.979% for two-stage, a gap of approximately 1.173 percentage points. Table 4 reports denominators and final warnings for this comparison.

The single-stage student has higher reference agreement in every reported action category: string–hui F1 is 37.511 versus 35.444, left-hand F1 is 57.917 versus 57.119, right-hand F1 is 47.697 versus 46.009, tone-type F1 is 65.629 versus 62.543, and ornament F1 is 20.929 versus 17.013 (Table 2). The two-stage pipeline has fewer final warnings, 2 versus 381, with rule violation rates of 0.194% versus 14.258%. These results show a trade-off: the current two-stage pipeline achieves greater pitch and runtime-rule consistency, while single-stage generation more closely reproduces the reference actions and ornaments. Performer assessment is needed to determine how these differences affect execution and musical expression.

Teacher trajectory construction. We compare our single-stage variant with a student trained on fixed-workflow trajectories. Each fixed trajectory contains generic reasoning, one real pitch-candidate query, and one edit constructed from the original expert annotation, without a teacher API rollout. Our teacher trajectories instead supervise the teacher's final committed plans after interaction. Both students are evaluated with diagnostic feedback and a 16-round stage limit on the same 48 scores using complete generated histories. The recipes differ in interaction traces, final targets, and some system/user prompts; this comparison evaluates the complete construction recipes rather than isolating reasoning supervision.

The fixed-workflow student achieves 100% protocol-valid coverage, but pitch accuracy is 26.778% on 13,556 evaluable events and MAE is 517.624 cents. The standard single-stage student attains 98.732% on 15,935 events and 7.880 cents MAE. On their 12,792 common evaluable events, accuracy is 26.931% versus 98.781% (Table 5). Thus, the large pitch gap persists on a shared event set rather than arising solely from different parseability.

The fixed-workflow model leaves 11,596 warnings across 988 units, including 9,926 pitch-mismatch warnings. Its string–hui F1 is 11.002 versus 37.511 for the standard single-stage student. However, its ornament F1 is higher (27.928 versus 20.929), as is technique distribution similarity (0.8401 versus 0.7618). Matching reference technique usage therefore does not ensure that those actions realize the input pitches. The results favor expert-guided trajectories for pitch consistency under the evaluated recipes, while leaving the separate roles of reasoning content, iterative feedback, prompt differences, and training length unresolved.

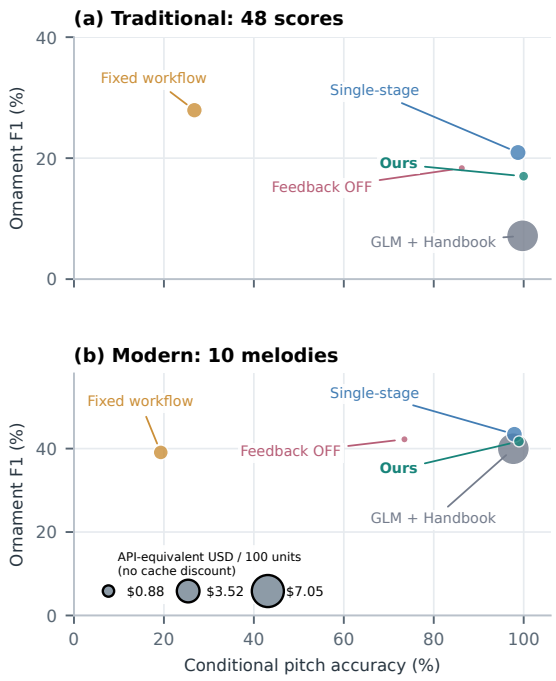


Figure 2: Quality and cost trade-offs on traditional scores (top) and modern melodies (bottom). All bubble areas use one linear scale for estimated API-equivalent cost per work unit; legend values are USD per 100 units. Estimates retokenize retained visible trajectories at public model-specific rates with no cache discount (Section 4.4). Pitch accuracy is conditional on resolvability; ornament F1 measures reference agreement, not musical expression.

4.4 Modern Melodies

We evaluate the two-stage system, single-stage student, fixed-workflow student, diagnostic-feedback-off variant, and GLM-5.3 ReAct + Handbook on the same ten modern melodies (91 work units) using complete generated score histories (Table 2, lower panel). All five methods produce protocol-valid outputs for every unit. Pitch is audited against the input melody; source-score jianzipu annotations provide reference actions for agreement metrics. The sealed references match all 91 public input IDs, hashes, and runtime inputs, and are used only for offline evaluation. These annotations represent one realization, not an assessment by the performers in our planned study. The independence of these melodies from all training and model-selection procedures remains to be verified, so this analysis is not yet a strictly held-out generalization benchmark.

Figure 2 shows why pitch fidelity and reference technique agreement should be examined jointly. The fixed-workflow model can reproduce ornaments while missing input pitches; our single-stage variant has higher ornament agreement than the full pipeline, whereas the latter has higher conditional pitch accuracy. Bubble areas add the same API-equivalent cost estimate to both datasets, using each method's final retained trajectories.

Inference cost. We distinguish recorded token usage from a reproducible API-equivalent cost estimate. For input tokens I, cache-hit input tokens H (a subset of I), and output tokens O, the token charge is

$$C = \left[r_{\text{in}}(I - H) + r_{\text{cache}}H + r_{\text{out}}O\right]/10^6, \quad (11)$$

where rates are USD per million tokens; cache creation and storage are additional when applicable. We use Together AI's public service prices checked on November 2, 2026: Qwen3.5-9B input/output rates are \$0.17/\$0.25, and GLM-5.3 input/cached-input/output rates are \$1.40/\$0.26/\$4.40.[1] Figure 2 uses model-specific public tokenizers and chat templates to retokenize the visible requests and responses retained with each final output. Qwen requests are saved verbatim; GLM inputs are reconstructed from saved user prompts and tool receipts with the production system prompt and fixed handbook. Server-side cache-hit counts were not retained, so we use an explicit no-cache-discount scenario ($H = 0$), not a claim of measured zero cache hits. These estimates exclude discarded runs, training, external tool execution, and unrecorded hidden reasoning or provider-only framing. Failed calls whose response usage was not saved are also excluded, not assigned zero cost; preflight context rejections are tracked separately. They describe retained visible-trajectory workload at public API rates, not complete historical usage, actual invoices, or local GPU expenses.

Of 1,528 events with a nonempty target pitch, 1,334 are auditor-resolvable for the two-stage system and 1,402 for the single-stage system, corresponding to 87.304% and 91.754% coverage. Conditional PitchAcc@50c is 98.951% and 97.932%, respectively; the proportion of all 1,528 target events both resolved and correct is 86.387% and 89.856%. On the 1,248 events resolvable for both, the corresponding accuracies are 99.920% and 98.157%. Full-score runtime replay yields 15 warnings across three work units for two-stage, versus

[1] https://www.together.ai/pricing; GLM rates also match https://docs.z.ai/guides/overview/pricing. These are provider list prices, not a manufacturer price for self-hosted adapters.

42 across 17 for single-stage. These are rule-based diagnostics, not independent performer judgments; the stage and retry budgets also differ between methods.

ReAct + Handbook attains 97.699% pitch accuracy on 1,347 auditor-resolvable events, with 16.035 cents MAE and 33 warnings across five units. Its string–hui F1, technique similarity, and ornament F1 are 36.627, 0.7118, and 39.961, respectively. As on traditional scores, the single-stage student's reference agreement can exceed that of the two-stage system despite lower conditional pitch accuracy. Disabling diagnostic feedback yields 73.485% pitch accuracy on 1,403 evaluable events and 472 final warnings across 78 units. On the 1,271 events resolvable for both ON and OFF, accuracy is 98.899% versus 73.879%. OFF's technique similarity (0.8623) exceeds the full system's, again showing that aggregate reference usage and pitch consistency capture different properties.

The fixed-workflow student obtains 19.298% pitch accuracy on 1,197 evaluable events, with 558.105 cents MAE and 1,060 warnings across 88 units. On the 1,119 events resolvable for both single-stage students, the teacher-trajectory student attains 98.749% versus 19.929% for fixed workflow. Thus, the large construction-recipe gap also occurs on modern melodies and persists on shared events. Prompt and training-trajectory differences still limit causal attribution to any individual construction step.

4.5 Expert Evaluation

A blinded performer study is planned with one or two guqin experts who will actually play selected scores. Each will rate two dimensions on a four-point scale: *performance feasibility*, covering string and fingering choices and continuity, and *idiomatic and overall musical expression*, covering the suitability of techniques and audible *yunwei*. Scale anchors and sampling will be fixed before assessment. Examples will be anonymized and presented in randomized order, with input scores available to raters. We will report rater experience and sampling, and inter-rater agreement if two experts assess the same examples. Expert scores and claims of superior playability or expression await this assessment.

5 Conclusion

GUQINAGENTframes jianzipu generation as executable, stateful plan editing. Its inverse trajectory construction turns private expert annotations into public tool-use supervision, supporting both single-stage generation and a two-stage pipeline for basic realization and idiomatic refinement. Both variants complete all 1,031 work units in the 48-score test. The full system achieves 99.981% pitch accuracy on auditor-resolvable events and leaves two runtime warnings, while our single-stage variant achieves higher reference agreement across fingering, tone type, and ornaments. On modern melodies, the full system also has higher conditional pitch accuracy, whereas single-stage generation resolves more target events and yields more verified-correct events overall. The fixed-workflow comparison shows that reproducing reference technique patterns alone does not ensure pitch consistency. Together, these results distinguish pitch fidelity, coverage, and reference agreement as complementary aspects of performance realization. Performer-rated feasibility and musical expression remain to be measured; the final conclusions will incorporate that evidence and the remaining comparisons.

6 Limitations, Ethics, and Resource Availability

The auditor is conservative and incomplete. Fine-grained left-hand reachability, timbre, and lineage-specific style require performer assessment or additional physical modeling. Inference feedback and final warning scoring share domain rules, so warning reduction measures rule consistency rather than independent musical quality. Generated notation also affects pitch resolvability; conditional accuracy must be read alongside coverage. The comparisons use different interaction budgets and trajectory-construction recipes, as specified in Section 4, and do not isolate every component's causal effect. The independence of the modern melodies from all development procedures remains to be verified. The digitized collection may not represent all historical sources or playing traditions. GUQINAGENTis intended to support performers and learners in preparing editable realizations. Performer evaluation and the GEPA comparison remain pending.

Open Science and Resource Availability. We will release the trained model, redacted training

trajectories, the test benchmark, and the executable runtime used by teacher generation and inference. The release will include score-level split definitions, event-indexed schemas, evaluation scripts, filtering manifests, and instructions for reproducing the benchmark. Private teacher prompts, unredacted teacher reasoning, and any source material that cannot be redistributed will not be included. The archival repository URL, model license, dataset license, and versioned release identifier will be added before submission: [TODO: repository URL and licenses].

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A Training and Inference Configuration

Training. All four students use LLaMA-Factory with the `qwen3_5` chat template, bitsandbytes 4-bit quantization, BF16 computation, and LoRA targeting all supported linear projections. The saved adapters have rank 8, $\alpha = 16$, dropout 0, and no bias adaptation. Their target modules are `q/k/v/o_proj`, `gate/up/down_proj`, `in_proj_qkv`, `in_proj_z`, `in_proj_b`, `in_proj_a`, and `out_proj`. All use a cosine learning-rate schedule, warmup ratio 0.03, learning rate 2×10^{-4}, three epochs, and gradient checkpointing. Prompt tokens are not supervised (`train_on_prompt=false`); assistant history is not masked (`mask_history=false`). Direct targets contain only the final JSON and explicitly disable thinking. Its 3,976 cached targets are fully supervised without truncation; this is not a claim about the longer tool trajectories.

Setting	Two-stage	Single-stage recipes
Training scores	153	153
Underlying work units	3,976	3,976
Training conversations	7,952	3,976
Training GPUs	2	1
Batch per GPU	1	1
Gradient accumulation	6	12
Effective batch	12	12
Training cutoff (tokens)	8,192	8,192
Epochs	3	3
Optimizer updates	1,989	996

Table 6: Verified training configurations. Single-stage recipes comprise trajectory-trained single-stage, fixed-workflow SFT, and Direct SFT. The two-stage adapter is trained jointly on both stage types; OFF reuses that adapter without retraining.

Inference. Local Qwen inference uses vLLM with bitsandbytes 4-bit weights, the corresponding LoRA adapter, a 16,384-token context, and an 8,192-token per-request output limit. Tool-based adapted conditions use attempt temperatures $[0.6, 0.3, 0.0]$, top-$p = 0.95$, top-$k = 20$, min-$p = 0$, presence penalty 0, repetition penalty 1, and seed 20261024. A round is one model response, which may request tools; it is not necessarily one tool call. The limits apply per attempt, not to the aggregate trajectory. Two-stage ON permits up to three Guqinizer passes; OFF disables diagnostic follow-ups and uses one. Direct uses temperature 0, thinking disabled, no tool schema, and exactly one generation per attempted work unit. Within a score, all conditions preserve their own generated history rather than substituting reference plans. Failed predecessors cause subsequent work units to be skipped under the strict-history protocol.

GLM-5.3 ReAct + Handbook uses the production single-stage prompt, a fixed 73-entry handbook, eight rounds, two attempts, and temperature 0.25. Its saved launch configurations do not record the effective output-token cap or API context limit. The client code defaults to 8,000 output tokens but permits an environment override; we therefore do not present that default as a verified historical cap. GEPA remains pending and has no reported final evaluation budget or score.

Artifact fingerprints. Table 7 identifies the exact training JSONL files and final adapter weight files (`adapter_model.safetensors`) by full SHA256. The same checkpoint is used for traditional and modern evaluation within each method. Training configuration files in `train/configs/` use the suffixes `two_stage_final_2gpu_bs12`, `single_stage_final_1gpu_bs12`, `fixed_reference_ablation_1gpu_bs12`, and `direct_sft_1gpu_bs12`, each prefixed by `qwen35_9b_lora_` and ending in `.yaml`. Their independent data directories and preprocessing manifests bind the tokenized caches to the source data, template, and cutoff. The fixed-workflow prompts partly differ from the trajectory-trained single-stage prompts; this comparison is not a strict single-variable ablation.

B Additional Action and Tone-Type Results

Table 8 reports event-level left- and right-hand fingering and tone-type F1, alongside tone-type distribution similarity. These complement the pitch, string–hui position, technique-distribution, ornament, and warning metrics in the main comparison. Tone similarity compares aggregate proportions of open, stopped, and harmonic notes and does not require these tone types to occur at the same events.

C API-Equivalent Inference Cost

Training recipe	Training JSONL SHA256	Adapter weights SHA256
Two-stage (ON / OFF)	6ea8329df61ade844389611d76ba5e32 15deba19f9a9aa6e7e0ae1d8c8d150b6 984779ea61a9c582328518bf8d4e0238	ecdee4141e31cabad373a7dba0fbcddc f0a03d15ff11f46bb41ca02ec510d0ad cdcc420e0b193cb1b6ad2520af13ae43
Single-stage	26bf31b8c705f927c1dfbd9d97b00f4a 72f664d51ca2aecb62e792d20043f8bf	b5549bac63cb8d4e3d7b0123daefb436 8447be00f62ad6a81c605b5d1cb59e7b
Fixed workflow	befe494c64d10c1a4b6f0e68c81ffc6a edba4abf12f1d2834bd079ea30f7ed1e	7149582b7caddec987e33ba9ee57ac29 30c0e8fcd8fb7387d86f66677f8596c2
Direct SFT	f50238b2a631db6948712b18d102f7e9	4327a693eced4f5ac8131dfcb27d0913

Table 7: Exact artifact identities. Concatenate the two lines in each hash cell without whitespace. Direct training is complete, but its test metrics are not yet reported.

Method	LH F1	RH F1	Tone F1	Tone Sim.
GLM + Handbook	43.609	40.835	54.555	**0.8439**
Fixed-workflow SFT	54.965	43.505	59.102	0.8320
GUQINAGENT, single-stage	**57.917**	**47.697**	**65.629**	0.8113
GUQINAGENT w/o feedback	56.599	45.659	61.670	0.7810
GUQINAGENT (Ours)	57.119	46.009	62.543	0.7743

Table 8: Additional structured action metrics on the same 48-score test set. F1 is a percentage; tone similarity is $1 - \mathrm{TVD}$ on $[0, 1]$. Higher values indicate closer agreement with reference annotations. All five conditions produce protocol-valid outputs for all 1,031 work units.

Method	LH F1	RH F1	Tone F1	Tone Sim.
GLM + Handbook	53.253	49.954	61.973	**0.8543**
Fixed-workflow SFT	52.970	46.019	65.576	0.7199
GUQINAGENT, single-stage	58.171	55.347	69.681	0.7963
GUQINAGENT w/o feedback	**59.350**	56.980	**70.049**	0.7643
GUQINAGENT (Ours)	57.255	**59.010**	67.663	0.7774

Table 9: Additional action metrics on ten modern melodies (91 work units), using source-score annotations as offline references. F1 is a percentage; tone similarity is on $[0, 1]$. All five methods have 100% protocol-valid coverage.

Method	Traditional	Modern
GLM + Handbook	7.046	7.012
Fixed workflow	1.703	1.539
single-stage	1.829	1.770
feedback OFF	0.374	0.370
GuqinAgent (Ours)	0.700	0.824

Table 10: Estimated API-equivalent USD per 100 work units, using the same visible-trajectory scope and no-cache-discount scenario in both datasets. These are retokenized estimates at public service rates, not historical invoices or self-hosting expenses. See Section 4.4 for the accounting definition.